Geometric Origin of the Fine-Structure Constant α — Trilogy (Papers 7, 8, 9)

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Overview of the trilogy: Observation → Structure → Physical Content Two elementary geometric quantities of the 4D unit ball satisfy the self-consistent equation

$$\alpha^{-1} = 137 + \tfrac{\pi^2}{2}\alpha \quad \Longleftrightarrow \quad \tfrac{\pi^2}{2}\alpha^2 + 137\alpha - 1 = 0$$

whose positive root agrees with the CODATA value at **8.7 ppb** precision (about 1/3000 of Eddington-style integer fitting deviation). This trilogy positions its geometric origin in three layers:

- **Paper 7 (Algebraic observation)**: discovery of the identity and establishment of numerical precision
- **Paper 8 (Mathematical correspondence)**: **structural isomorphism** with Wilson standard lattice gauge theory on the 4D chain complex
- **Paper 9 (Physical content)**: from the area dimensionality of α, only 2D face vibration modes contribute. $\pi^2/2$ is their position phase space measure, and $(\pi^2/2)\alpha$ is the collective zero-point contribution. Physically isomorphic to Wilson plaquette action.

Common caveat Neither paper claims a first-principles derivation of α. Paper 9's scope is only the Thomson limit $\alpha^{-1}(Q^2 \to 0)$; the high-energy running is deferred to standard QED as future work.

Paper 7: α Identity (Algebraic Observation)

Direct counting of $N(1) = 137$ Unit cubes centered on integer points fully contained in the 4-ball of radius $R = 3$, enumerated by absolute-value patterns $(s_1, \ldots, s_4)$ satisfying $\sum_i (|c_i| + 1/2)^2 \le 9$:

Pattern	Count	Pattern	Count
$(0,0,0,0)$	1	$(1,1,1,1)$	16
$(1,0,0,0)$	8	$(2,0,0,0)$	8
$(1,1,0,0)$	24	$(2,1,0,0)$	48
$(1,1,1,0)$	32	Total	**137**

$V_4(1) = \pi^2/2$ standard formula $V_n(R) = \pi^{n/2}/\Gamma(n/2+1) \cdot R^n$ with $n = 4, R = 1$ gives $V_4(1) = \pi^2/2 \approx 4.9348$.

Numerical verification (pocket calculator)

$V_4(1) = \pi^2/2$	≈ 4.9348022
$N(1)$	$= 137$
$D = 137^2 + 2\pi^2$	≈ 18788.7392
α (high precision)	$\approx 7.29735194 \times 10^{-3}$
α^{-1} (predicted)	≈ 137.0360110
α^{-1} (CODATA 2018)	$= 137.0359991\ldots$
Relative error	$\mathbf{8.7 \times 10^{-8}}$

Differences from the Eddington trap (1) Not a single integer-fitting formula but a **self-consistent equation** including α itself. (2) Both 137 and $\pi^2/2$ are **directly derived from independent 4D geometry**. (3) The perturbative form $1 = 137\alpha + V_4(1)\alpha^2$ is analogous to QFT renormalisation structure.

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Paper 8: Wilson Isomorphism (Mathematical Correspondence)

Chain complex on $\mathbb{Z}^4$ Standard hierarchy:

- C_0 vertex / C_1 link / C_2 plaquette
- C_3 3-cell / C_4 4-cell (tesseract)
- Hypercubic symmetry group B_4 (order 384) acts equivariantly

Schläfli duality connects the two formalisms Tesseract $\{4,3,3\}$ and 16-cell $\{3,3,4\}$ are self-dual:

Λ side (Wilson)	Λ^* side (Kihara)
C_1 link (gauge field U_ℓ)	C_3 3-cell
C_2 plaquette (action)	C_2 plaquette (self-dual)
C_4 4-cell	C_0 site (packing center)

Discrete Stokes ($\partial^2 = 0$) common to both:

- Wilson: plaquette sum → boundary Wilson loop
- Kihara: bulk packing → boundary $\Delta(R)$

Main theorem (Structural correspondence) Wilson configuration $\mathcal{C}_W$ and Kihara configuration $\mathcal{C}_K$ are **structurally isomorphic as chain complexes** on the 4D Euclidean lattice via a B_4-equivariant chain map D (Schläfli duality).

Chain-complex reading of inside/outside decomposition $1 = 137\alpha + V_4(1)\alpha^2$ is the bulk-boundary normalisation:

- 137α: C_4 bulk (137 cells) tree-level
- $V_4(1)\alpha^2$: C_3 boundary self-energy correction

Consequence: Bidirectional tool sharing Wilson strong-coupling expansion ↔ inclusion-exclusion, Monte Carlo ↔ numerical verification, RG ↔ continuum limit, Lagrange–Jacobi $r_4(N) = 8\sigma(N)$ ↔ partition function.

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Paper 9: Physical Interpretation via 2D Surface Vibration Modes (Physical Content)

Core: $\pi^2/2$ is read not as $V_4(1)$ but as the position phase space measure of 2D face modes Although α is dimensionless, since $\sigma_T \propto \alpha^2$, it is physically **area-dimensional**. Therefore, among 4D space vibration modes, only **2D face modes** can contribute to α under isotropic averaging (0D/1D/3D/4D vanish by directional cancellation; 2D area is scalar and survives). The integral of position degrees of freedom for placing 2D faces of 137 hypercubes in the 4D unit ball is

$$\int_{B_4(1)} dV = \frac{\pi^2}{2}$$

which coincides numerically with $V_4(1)$. The collective zero-point vibration $\frac{1}{2}\hbar\omega$ of each 2D face mode produces the $(\pi^2/2)\alpha \approx 0.036$ drift. This has **identical structure** to Wilson plaquette action (physical content of Paper 8 isomorphism).

Dimensional analysis of α Quantities through which α participates physically appear as α^2:

- Thomson: $\sigma_T = (8\pi/3)r_e^2 \propto \alpha^2 \cdot (\text{length})^2$
- Rutherford: $d\sigma/d\Omega \propto \alpha^2 \cdot (\text{length})^2$
- Classical electron radius: $r_e = \alpha \cdot \hbar/(mc)$

$\therefore$ α is a "geometric square root of area"—a quantity coupling to 2D structures.

Dimensional selection of vibration modes

Dim.	Property	Avg. result
0D vertex	vector position	zero
1D edge	has direction	zero
2D face	**scalar quantity**	**survives**
3D solid	pseudoscalar	zero
4D 4-cell	pseudoscalar	zero

Reading the physical meaning of $\pi^2/2$

Old (W7)	4D unit ball volume $V_4(1)$
New (W9)	**2D face mode position phase space measure**

The two are numerically identical but physically different. Since α couples to 2D face modes, the volume of their position phase space necessarily appears as the geometric coefficient of the coupling.

Correspondence with Wilson plaquette action (physicalised)

Wilson lattice	W7 self-consistency (W9 reading)
$\beta = 1/g^2$	α^{-1}
plaquette (2-form)	2D face vibration mode
plaquette count $\sum_p$	$\pi^2/2$ (position phase space)
$\beta \sum_p \mathrm{Re}\,\mathrm{tr}(U_p)$	$(\pi^2/2)\alpha$
Cell (4D hard core)	137
Total action	$\alpha^{-1} = 137 + (\pi^2/2)\alpha$

Consequence: physical content of W8 isomorphism The W7 self-consistent equation is the **realisation** of the plaquette action structure of Wilson lattice gauge theory on the 4D ball geometry.

Scope and limitations This paper addresses only $\alpha^{-1}(Q^2 \to 0) = 137.036$ (Thomson limit). The running to $\alpha^{-1}(M_Z) \approx 127.95$ is left to standard QED:

- Layer 1: 137 (integer, geometric invariant, Paper 6)
- Layer 2: $(\pi^2/2)\alpha \approx 0.036$ (**this paper**)
- Layer 3: $\Delta\alpha_{SM}(Q^2)$ (standard QED)

Theoretical lower bound on observed α 2D face mode zero-point vibrations have finite distribution width from the uncertainty principle. α is not a sharp point but the center of a distribution.

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Common open problems (trilogy) • Geometric identification of the 8.7 ppb residual (α^3-order correction; partially observed in Paper 7 Supplement, Concept DOI 10.5281/zenodo.19933729) • Privileged role of R=3 (why $N(1)$ corresponds to electron) • First-principles derivation of mechanism candidates (vacuum polarisation / boundary degrees / Higgs longitudinal mode) • Geometric reformulation of high-energy running (Paper 9 §8, future task) • Generalisation to other gauge groups (SU(2), SU(3)) • Projection to physical (3+1)-dim spacetime • Generational mass hierarchy out of scope

What the trilogy does NOT claim (common) First-principles derivation of α / 4+1-dim physical spacetime / replacement of Eddington-style predictions / new observable signatures / independent derivation of $\alpha(M_Z)$ running (deferred to standard QED)

Zenodo · note · Zenn Resources

	Paper 7 (α identity)	Paper 9 (2D mode interpretation)
note JA	n19cc13927c51	nd7ae96120b66
note EN	n01d98237ddae	nf9b18cb9a254
Zenn article	alpha-identity-4d-geometry	paper9-2d-mode-drift
	Paper 8 (Wilson isomorphism)	Paper 7 Supplement (2nd-order)
note JA	n87df7a4977e7	ne1dc24c07455
note EN	n9c37c6f99a0a	n268fa839f6c4
Zenn article	alpha-isomorphism-lattice-gauge	paper7-supplement-second-order-observation

Source code (GitHub) `https://github.com/WurabeSeiji/ai-chat-logs-open`

Related programs BH Thermodynamics Program (core 6 papers): Zenn Phase Equation Series (W1–W11): Zenn Working Paper (6D extension): Zenodo